although I could not hazard a guess as to what the details might be, nor by what tortuously indirect routes the influence might travel.

Survival machines of different species influence each other in a variety of ways. They may be predators or prey, parasites or hosts, competitors for some scarce resource. They may be exploited in special ways, as for instance when bees are used as pollen carriers by flowers.

Survival machines of the same species tend to impinge on each others' lives more directly. This is for many reasons. One is that half the population of one's own species may be potential mates, and potentially hard-working and exploitable parents to one's children. Another reason is that members of the same species, being very similar to each other, being machines for preserving genes in the same kind of place, with the same kind of way of life, are particularly direct competitors for all the resources necessary for life. To a blackbird, a mole may be a competitor, but it is not nearly so important a competitor as another blackbird. Moles and blackbirds may compete for worms, but blackbirds and blackbirds compete with each other for worms *and* for everything else. If they are members of the same sex, they may also compete for mating partners. For reasons that we shall see, it is usually the males who compete with each other for females. This means that a male might benefit his own genes if he does something detrimental to another male with whom he is competing.

The logical policy for a survival machine might therefore seem to be to murder its rivals, and then, preferably, to eat them. Although murder and cannibalism do occur in nature, they are not as common as a naïve interpretation of the selfish gene theory might predict. Indeed Konrad Lorenz, in *On Aggression*, stresses the restrained and gentlemanly nature of animal fighting. For him the notable thing about animal fights is that they are formal tournaments, played according to rules like those of boxing or fencing. Animals fight with gloved fists and blunted foils. Threat and bluff take the place of deadly earnest. Gestures of surrender are recognized by victors, who then refrain from dealing the killing blow or bite that our naïve theory might predict.

This interpretation of animal aggression as being restrained and formal can be disputed. In particular, it is certainly wrong to condemn poor old *Homo sapiens* as the only species to kill his own kind, the only inheritor of the mark of Cain, and similar melodramatic charges. Whether a naturalist stresses the violence or the restraint of animal